

Principles of Economics

Class 6: Supply and demand - Perfectly competitive markets and price-takers

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Market power vs Price-takers

The supply curve

The competitive equilibrium

Comparative statics

Comparative Statics of Taxation

Summary

1. Market power vs Price-takers

- ▶ With **market power**, the firm sets its price:
 - ▶ Constrained by demand and competition: higher prices, fewer sales.
- ▶ But competition often makes firms **price-takers**:
 - ▶ They take the market price as given: no market power.
- ▶ A firm is a price-taker if it can only sell at the market price (or less):
 - ▶ it **cannot influence the market price**;
 - ▶ Charge more and it sells nothing: **consumers buy from others**.
- ▶ This requires a very competitive environment:
 - ▶ **Many** suppliers with capacity to meet demand;
 - ▶ **Identical products**: the consumer does not care who sells.

- ▶ Does being a price-taker leave the firm **with no decisions**?
 - ▶ No! Given the price, **it only chooses the quantity**.
- ▶ **Two important questions:**
 - ▶ Does firm behaviour differ from the market-power case?
 - ▶ Does competition improve market outcomes?
- ▶ Today: profit maximisation for price-taking firms.
- ▶ Then: the model of perfect competition.

Recall profit maximisation

Profit-maximising quantity and price

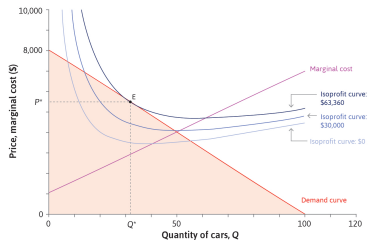
► Markup on the last unit sold: $P^* - MC(Q^*)$.

► Slope of demand (MRT): $\frac{\Delta P}{\Delta Q} = \frac{P}{Q} \frac{1}{\varepsilon_P}$

► Slope of the isocost (MRS): $\frac{P-MC}{Q}$

► Hence the “markup” formula:

$$\frac{P - MC}{Q} = \frac{1}{\varepsilon_P}$$



Price-taking firms

Equilibrium price

- ▶ For a price-taker, the firm's **feasible set** (demand curve) is **flat**:
 - ▶ What is the slope of a horizontal line?
 - ▶ Zero.
 - ▶ Where is the **slope of the isoprofit zero**?

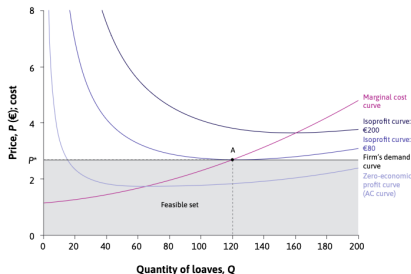
- ▶ Slope of the isoprofit
$$= \frac{P - MC}{Q}$$

- ▶ Therefore,

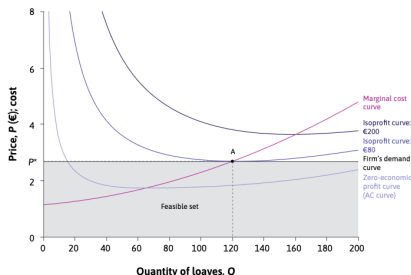
$$0 = \frac{P^* - MC}{Q} \Leftrightarrow P^* = MC$$

with P^* set by the market.

- ▶ Price-taking implies price equal to marginal cost.



- ▶ With $P^* = 2.35$, **can the firm do better** selling more than 120?
- ▶ Would the firm want to sell $Q = 65$?
 - ▶ No: at $Q = 65$ marginal cost (MC) crosses a lower isoprofit line.
 - ▶ With $P^* > MC$, **the firm gains by increasing the quantity.**
- ▶ What would happen with $Q = 160$?
 - ▶ No: at $Q = 160$ nobody buys at that price (roughly 4).
 - ▶ With $P^* < MC$, **the firm gains by reducing the quantity.**

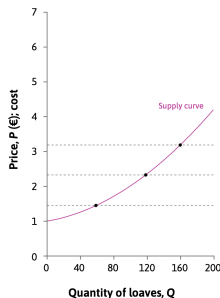
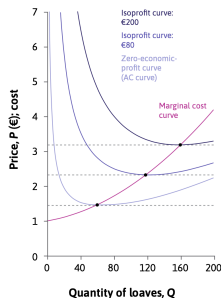


2. The supply curve

Price-taking firms

Equilibrium price

- ▶ At each price, the firm supplies the quantity that maximises profit. ($MC = P$)
- ▶ Quantity supplied by firm (i): $Q_i^s = S_i(P)$
- ▶ The market price then fixes the **equilibrium quantity of the market and of each firm**.



Price-taking firms

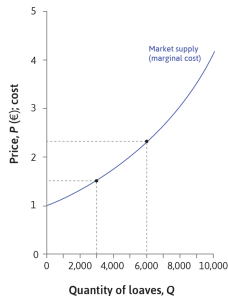
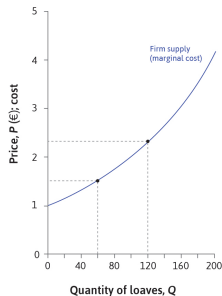
Market supply function

- ▶ Market supply: the sum of the quantities of the n firms at each price:

$$S(P) = S_1(P) + S_2(P) + \dots + S_n(P)$$

- ▶ With **50 identical firms**:

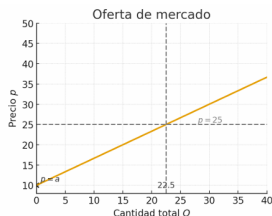
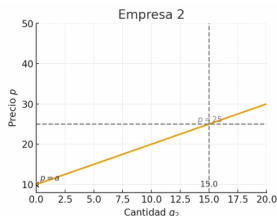
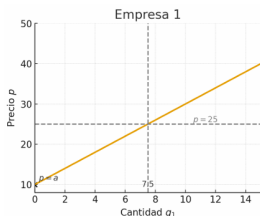
$$S(P) = 50 \times S_i(P)$$



Competitive equilibrium

Market supply function

- ▶ Firm 1, $MC_1 = 10 + 2Q_1$.
- ▶ Firm 2, $MC_2 = 10 + 2Q_2$.
- ▶ Firm 3, $MC_3 = 10 + 2Q_3$.
- ▶ Market supply $P(Q) = 10 + 2/3Q$



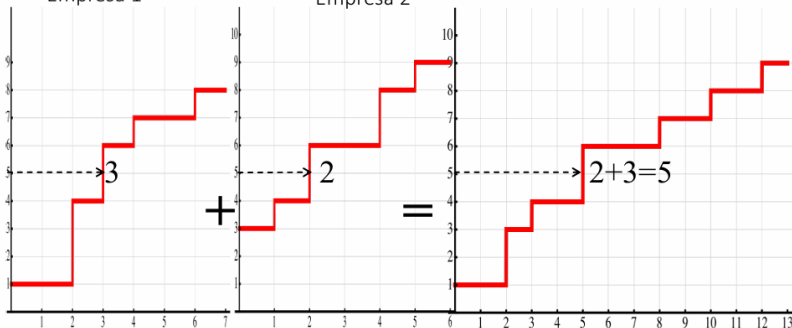


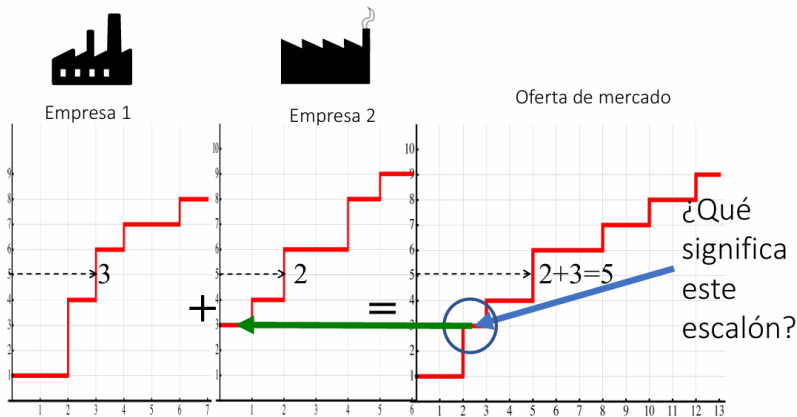
Empresa 1



Empresa 2

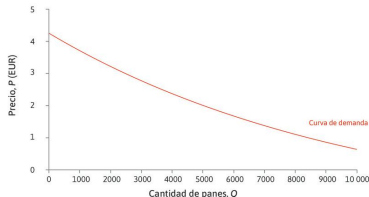
Oferta de mercado





3. The competitive equilibrium

- ▶ The market price comes from the **interaction of two forces**:



- ▶ **Demand curve** $D(P)$: total quantity consumers buy at each price;
- ▶ Normally decreasing in price:

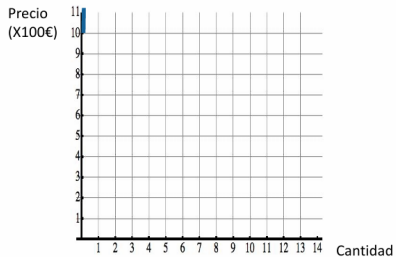
$$D(P) = A - BP$$

where $A, B > 0$

- ▶ It is buyers' **maximum willingness to pay (WTP)**.

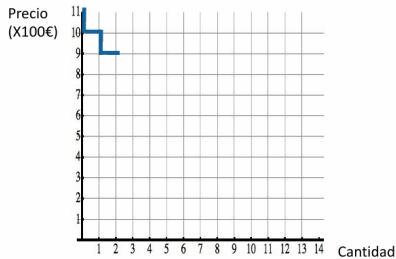
Example

Consumidores:



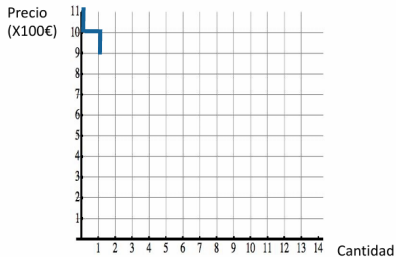
Example

Consumidores:



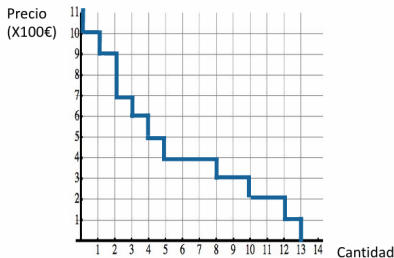
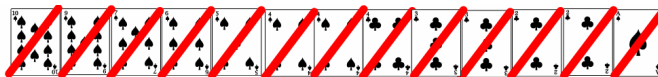
Example

Consumidores:



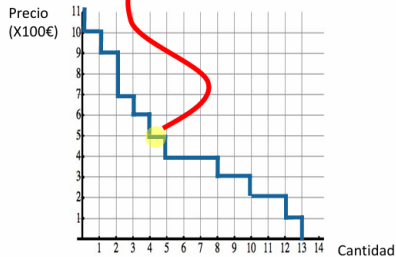
Example

Consumidores:

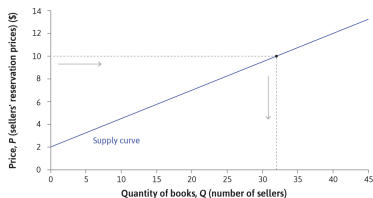


Example

Consumidores:



- ▶ The price each firm takes as given comes from the **interaction of two forces**:



- ▶ **Market supply curve $S(P)$** : total quantity that all firms want to sell;
- ▶ Normally increasing in price;

$$S(P) = C + DP$$

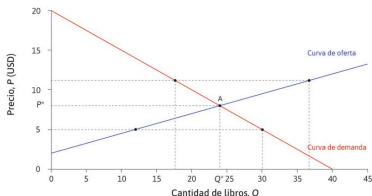
with $D > 0$.

- ▶ It is marginal cost, also called **willingness to accept (WTA)**.
- ▶ Note: sellers differ in their **reservation prices**: above the market price they **stay out of the market**.

Competitive equilibrium

Equilibrium price

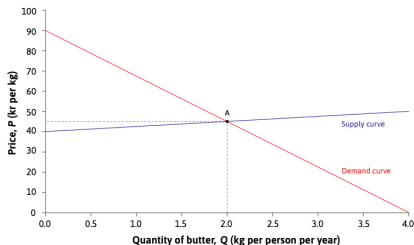
- ▶ The price each firm takes as given comes from the **interaction of two forces**:



- ▶ In equilibrium (**market clearing**), P^* equates supply and demand.
- ▶ Any other price is not an **equilibrium**: there are gains from deviating:
 - ▶ If $P > P^*$, **excess supply**: some seller gains by cutting the price
 - ▶ If $P < P^*$ there is **excess demand**.
- ▶ Products are identical: buyers accept any seller.

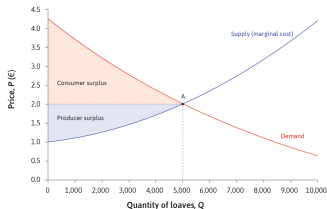
- **Example 1:** Demand for butter in Denmark. Suppose that demand and supply are given by:

$$D(P) = 4 - \frac{2}{45}P \text{ and } S(P) = -16 + \frac{2}{5}P$$



- **The competitive equilibrium** determines: $D(P) = S(P)$
 $\Leftrightarrow 4 - \frac{2}{45}P = -16 + \frac{2}{5}P$
 $\Leftrightarrow P^* = 45$
- Going back to demand or supply,
 $Q = D(45) = 4 - \frac{2}{45} \times 45 = 2$

- ▶ In this equilibrium **all the gains from trade are exhausted**:
 - ▶ There is **no efficiency loss**;
 - ▶ Total surplus is at its **maximum**.
 - ▶ CS: total difference between WTP and P^*
 - ▶ PS: total difference between WTA and P^*
- ▶ But Pareto efficiency is **not equity**:



- ▶ Here the **consumer surplus (CS)** is larger than the **producer surplus (PS)**;
- ▶ The **distribution of the surplus** depends on the shape of the curves.

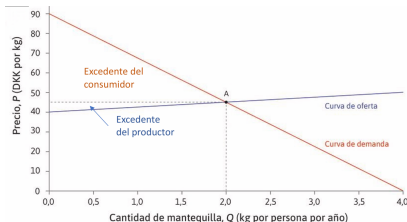
- ▶ The equilibrium in competitive markets is **Pareto efficient**

Consumer and producer surplus

- **Example 2:** Demand for butter in Denmark. Suppose that demand and supply are given by:

$$D(P) = 4 - \frac{2}{45}P \text{ and } S(P) = -16 + \frac{2}{5}P$$

- We know from before that: $(Q^*, P^*) = (2, 45)$



- CS is the difference between WTP and the market price:

$$CS = \frac{1}{2}(90 - P^*) \times Q^* = \frac{1}{2}(90 - 45) \times 2 = 45$$

- PS is the difference between the market price and WTA/MC:

$$PS = \frac{1}{2}(P^* - 40) \times Q^* = \frac{1}{2}(45 - 40) \times 2 = 5$$

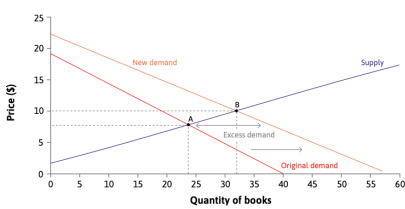
- $TS = CS + PS = 45 + 5 = 50$

4. Comparative statics

Changes in equilibrium

Exogenous demand and supply shocks

- ▶ **Exogenous changes shift** the demand or the supply curve.
- ▶ Buyers and sellers adjust until a new equilibrium is reached.
- ▶ Second-hand books: the new school year pushes students **to buy more**.



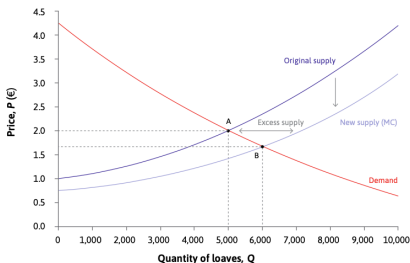
- ▶ **Demand rises at every price** (the curve shifts right);
- ▶ At the initial price there is **excess demand**;
- ▶ Both sides move along their curves to the new equilibrium:
 $Q_1 > Q_0$ and $P_1 > P_0$

- ▶ **Higher demand with unchanged supply: more quantity and a higher price!**

Changes in equilibrium

Exogenous demand and supply shocks

- ▶ Bread market: **technology improves and bakeries become more productive.**



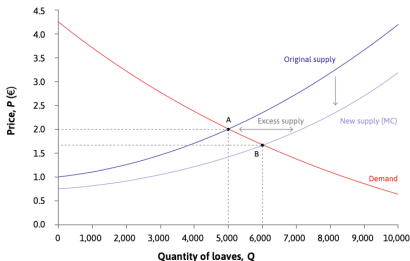
- ▶ **Supply rises at every price** (the curve shifts right);
- ▶ At the initial price there is **excess supply**;
- ▶ Both sides move along their curves to the new equilibrium:
 $Q_1 > Q_0$ and $P_1 < P_0$

- ▶ **Higher supply with unchanged demand: more quantity and a lower price!**

Changes in equilibrium

Exogenous demand and supply shocks

- ▶ Bread market: **new bakeries open in the area.**
- ▶ This is known as **market entry**.



- ▶ If new firms enter the market, **supply shifts to the right.**
- ▶ The entry of new competitors is **more likely** when:
 - ▶ Firms currently in the market have **positive economic rents**;
 - ▶ And the entry costs **are not too high**.
 - ▶ (The incentive ends when firms in the market have zero economic rent).

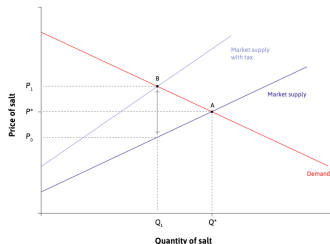
- ▶ **More competition with unchanged demand: more quantity and a lower price!**

5. Comparative Statics of Taxation

Changes in equilibrium

Taxes and market distortions

- ▶ **Taxes fund the government** (and redistribute).
- ▶ A tax shifts supply or demand: a higher price at every quantity.
- ▶ Suppose we are in the salt market:
 - ▶ The government imposes a sales tax of 30% per kilo.
 - ▶ Firms collect it and hand it to the government.
- ▶ **What is the impact on the equilibrium?**



- ▶ For the firm the tax is higher marginal cost;
- ▶ So, **supply shifts to the left**;
- ▶ The **new equilibrium** is such that:
 $Q_1 < Q^*$ and $P_1 > P^*$
- ▶ **Tax per unit:** $t = P_1 - P_0$

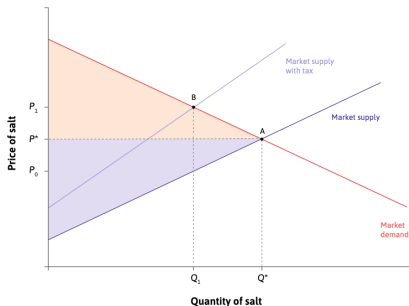
- ▶ Other taxes exist (income, for example), but here we mean a sales tax (like VAT)

Changes in equilibrium

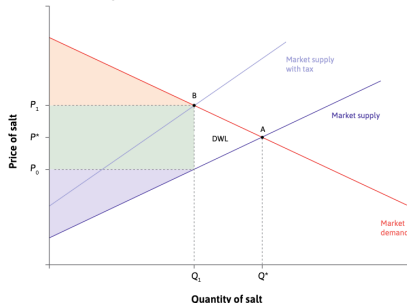
Taxes and market distortions

- ▶ Against the initial competitive equilibrium, **the tax cuts the gains from trade**:
- ▶ It generates a deadweight loss!

Initial equilibrium



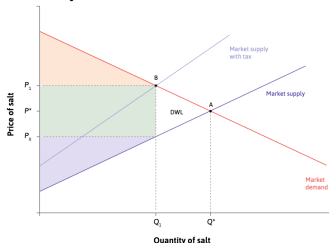
Equilibrium after the tax



- ▶ With a sales tax: CS decreases, PS decreases, and since $DWL > 0$, TS decreases.

- ▶ The fall in total surplus grows with the elasticity of demand (how sensitive consumers are to price);

Equilibrium after the tax



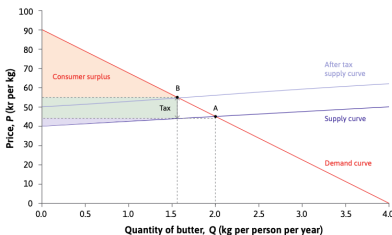
- ▶ The government takes part of the surplus: that is its **tax revenue**.
- ▶ Though the tax is levied on producers, **consumer surplus** also falls;

- ▶ **Tax incidence** reveals who bears the cost of the tax:
 - ▶ Usually the less elastic side bears more of the burden. (essential goods versus non-essential goods)
- ▶ The **welfare effect** depends on what the government does with the revenue (public goods, for example).

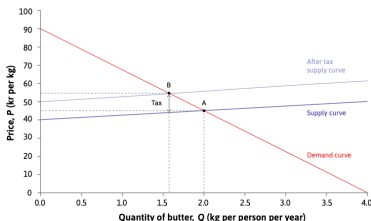
Competitive equilibrium

Changes in equilibrium: taxes

- ▶ Taxes can also change **consumer behaviour**:
 - ▶ E.g.: sugar taxes, environmental taxes, etc.
- ▶ **Denmark, 2011: a per-kg tax on saturated fats; butter rose 22%.**
- ▶ **Consumption of butter and related products fell by 15-20%.**
- ▶ **It was repealed: collecting it cost too much.**



- ▶ **Example 3:** Demand for butter in Denmark. Suppose that demand is given by $D(P) = 4 - \frac{2}{45}P$, and supply by $S(P) = \frac{2}{5}P - 16$. The government imposes a tax that increases producers' costs by 10 kroner per kg of butter.
- ▶ We know from before that: $(Q^*, P^*) = (2, 45)$, what is the **effect of the tax**?



- ▶ Supply after the tax is lower:

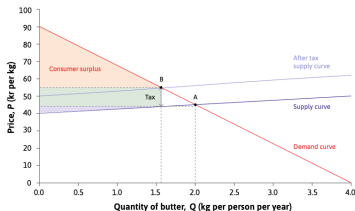
$$S(P)^{Tax} = \frac{2}{5}P - 20$$

- ▶ New equilibrium:

$$D(P) = S(P)^{Tax} \Rightarrow (Q^{**}, P^{**}) = (1.6, 54)$$

- ▶ The tax reduced the quantity and increased the price in the new equilibrium!

- ▶ **Example 3:** Demand for butter in Denmark. Suppose that demand is given by $D(P) = 4 - \frac{2}{45}P$, and supply by $S(P) = \frac{2}{5}P - 16$. The government imposes a tax that increases producers' costs by 10 kroner per kg of butter.
- ▶ What happens to **CS** and **PS**?

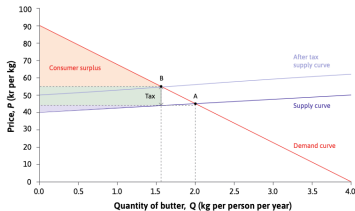


- ▶ We know from before that:
 $CS = 45$, $PS = 5$ and $TS = 50$
- ▶ Now:
 - ▶ $CS_{Tax} = \frac{1}{2}(90 - 54) \times 1.6 = 28.8$
 - ▶ Gov. revenue $= 10 \times 1.6 = 16$
 - ▶ $PS_{Tax} = \frac{1}{2}(44 - 40) \times 1.6 = 3.2$
 - ▶ $TS_{Tax} = 28.8 + 16 + 3.2 = 48$
- ▶ Therefore:
 $DWL = TS - TS_{Tax} = 50 - 48 = 2$

- ▶ **DWL** measures the efficiency loss due to the tax.

- ▶ **Example 3:** Demand for butter in Denmark. Suppose that demand is given by $D(P) = 4 - \frac{2}{45}P$, and supply by $S(P) = \frac{2}{5}P - 16$. The government imposes a tax that increases producers' costs by 10 kroner per kg of butter.

- ▶ **Namely:**



- ▶ The **consumer surplus** shrinks:
 $CS = 45$ and $CS_{Tax} = 28.8$
Therefore, $\Delta CS = 16.2$
- ▶ Therefore the **producer surplus**:
 $PS = 5$ and $PS_{Tax} = 3.2$
Therefore $\Delta PS = 1.8$

- ▶ Because demand is less elastic!
(a more vertical curve needs a bigger price change to move quantity)

6. Summary

- ▶ **Properties of a perfectly competitive market:**
 - ▶ The good or service traded is **homogeneous**;
 - ▶ **Many potential buyers and sellers**, acting independently;
 - ▶ There are **no barriers** to entering or leaving the market;
 - ▶ **Price information** is easily **accessible** to buyers and sellers.
- ▶ As a result:
 - ▶ **Law of one price**: a single price that clears the market (demand = supply)
 - ▶ Buyers and sellers are **price-takers**;
 - ▶ **All potential gains from trade** are realised.
- ▶ **Is all this realistic?**
 - ▶ It is a **benchmark**: few real markets meet every assumption.

► The two market structures, side by side:

Price-setters (e.g.: Monopoly)	Price-takers (Perfect Competition)
$MC < \text{Price}$	$MC = \text{Price}$
Deadweight losses (Pareto inefficiency)	No deadweight losses (can be Pareto efficient)
Owners receive economic rents both in the short and long run	No economic rents in the long run
Firms advertise their unique product	Little spending on advertising
Firms invest in R&D, seek to prevent copying	Little incentive for innovation

Today's class, section by section:

1. **Market power vs price-takers** → given the price, the firm only chooses the quantity.
2. **The supply curve** → $MC = P$ is the firm's supply; adding firms gives the market's.
3. **The competitive equilibrium** → demand equals supply and every gain from trade is exhausted.
4. **Comparative statics** → a shock shifts one curve and the equilibrium moves along the other.
5. **Comparative statics of taxation** → the tax raises the price, cuts the quantity and creates a deadweight loss.
6. **Price-takers vs price-setters** → $MC = P$ against $MC < P$, no rents in the long run against rents.

Key concepts: price-taker, market supply, competitive equilibrium, consumer and producer surplus, tax incidence, deadweight loss, perfect competition.